Flight calculus, in plain language

Hypothetical commercial flight: Los Angeles to New York.

1 / Name the quantities and their changes

Calculus connects how fast something is changing with how much has built up.

“Elapsed time” means time since takeoff; no actual flight data are used here.

Variable

A variable is a quantity whose value can change.

During our flight, elapsed time, altitude, fuel remaining and speed are variables.

Function

A function is a rule that gives one output for each input.

For our chosen flight, entering an elapsed time into the altitude record gives one altitude.

Independent and dependent variables

An independent variable is a chosen input, while a dependent variable is an output we follow as that input changes.

We choose elapsed time as input and follow altitude or fuel remaining as output.

Rate of change

A rate of change says how much one quantity changes compared with another.

Fuel-burn rate tells us how much fuel is used per unit of elapsed time.

Slope

Slope compares a vertical change on a graph with a horizontal change.

On an altitude-versus-elapsed-time graph, slope is vertical speed, not how quickly the airplane travels across the ground.

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2 / Find what is happening at one instant

Instantaneous versus average

An instantaneous rate describes one moment, while an average rate describes a whole interval.

Our average ground speed is total traveled path distance divided by elapsed time, while an instantaneous reading estimates ground speed at one moment.

Limit

A limit is the value a quantity approaches as we narrow in on a chosen situation.

Using shorter time intervals around one flight moment lets us approach its rate without ever dividing by a zero-length interval.

Derivative

A derivative is the exact instantaneous rate in an ideal smooth mathematical model.

The altitude derivative is zero in level cruise even though the airplane may be moving quickly across the ground.

Differentiation

Differentiation is the process of finding a derivative.

Differentiating our modeled altitude using elapsed time gives modeled vertical speed.

Tangent line

A tangent line matches a smooth curve’s direction at one chosen point.

Its slope on our altitude-time graph gives modeled vertical speed at that instant, not the triangle ratio called tangent.

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3 / Add up what happens along the way

Integral

An integral adds a continuously varying quantity over an interval to find a total.

The integral of our fuel-burn rate over elapsed time gives total fuel used, not fuel remaining.

Integration

Integration is the process of finding an integral.

Integrating ground speed over elapsed time gives distance along the flight’s traveled path, not the straight-line distance between its endpoints.

Area under a curve

For a positive rate-versus-time graph, area under the curve represents the accumulated total.

On our fuel-burn-rate graph the area means fuel used, whereas area under an altitude-time graph does not mean distance traveled.

Initial condition

An initial condition is a known starting value used to anchor a model.

Our model could start with a known fuel amount and subtract accumulated fuel use to find what remains.

Differential equation

A differential equation connects a changing quantity with one or more of its rates of change.

A simple flight-fuel model says that the rate of change of fuel remaining is the negative of the fuel-burn rate.

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4 / Describe motion and make sensible choices

Velocity

Velocity describes speed together with direction.

Our airplane’s velocity includes both its ground speed and its direction of travel.

Acceleration

Acceleration is a change in velocity, whether speed, direction or both change.

Our airplane accelerates during a turn even if its speed stays constant.

Maximum and minimum

A maximum is a highest value and a minimum is a lowest value within the range being considered.

For this flight, we could ask for the maximum altitude reached or the minimum fuel remaining over the modeled journey.

Optimization

Optimization means choosing the best available result while respecting stated limits.

A fuel-saving flight plan must still respect safety, weather and arrival requirements rather than simply seek the fastest trip.

Model

A model is a simplified description that helps us reason about a real situation.

Our smooth flight curves are models, while real instrument readings are measurements with uncertainty.

A conversation starter: “The derivative tells me the rate right now; the integral tells me how much has added up.”